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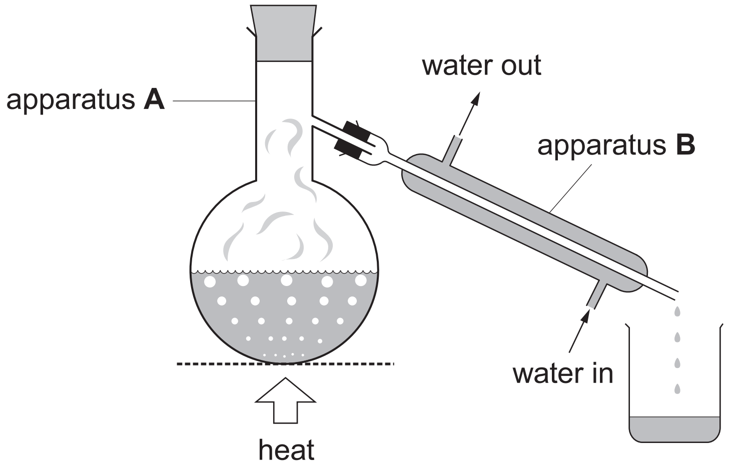
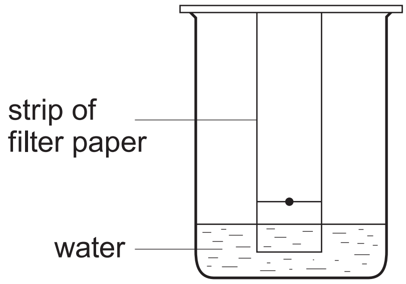
3. (a) The diagrams show two methods used to separate mixtures.



Method **1**



Method **2**

(i) I. Name the changes of state happening in apparatus **A** and apparatus **B** when water is separated from salt in Method **1**. [2]

Apparatus **A** ....

Apparatus **B** ....

II. Name this method of separation. [1]

....

(ii) Explain how an orange dye is separated into a red spot and a yellow spot on the filter paper in Method **2**. [2]

....

....

....

....

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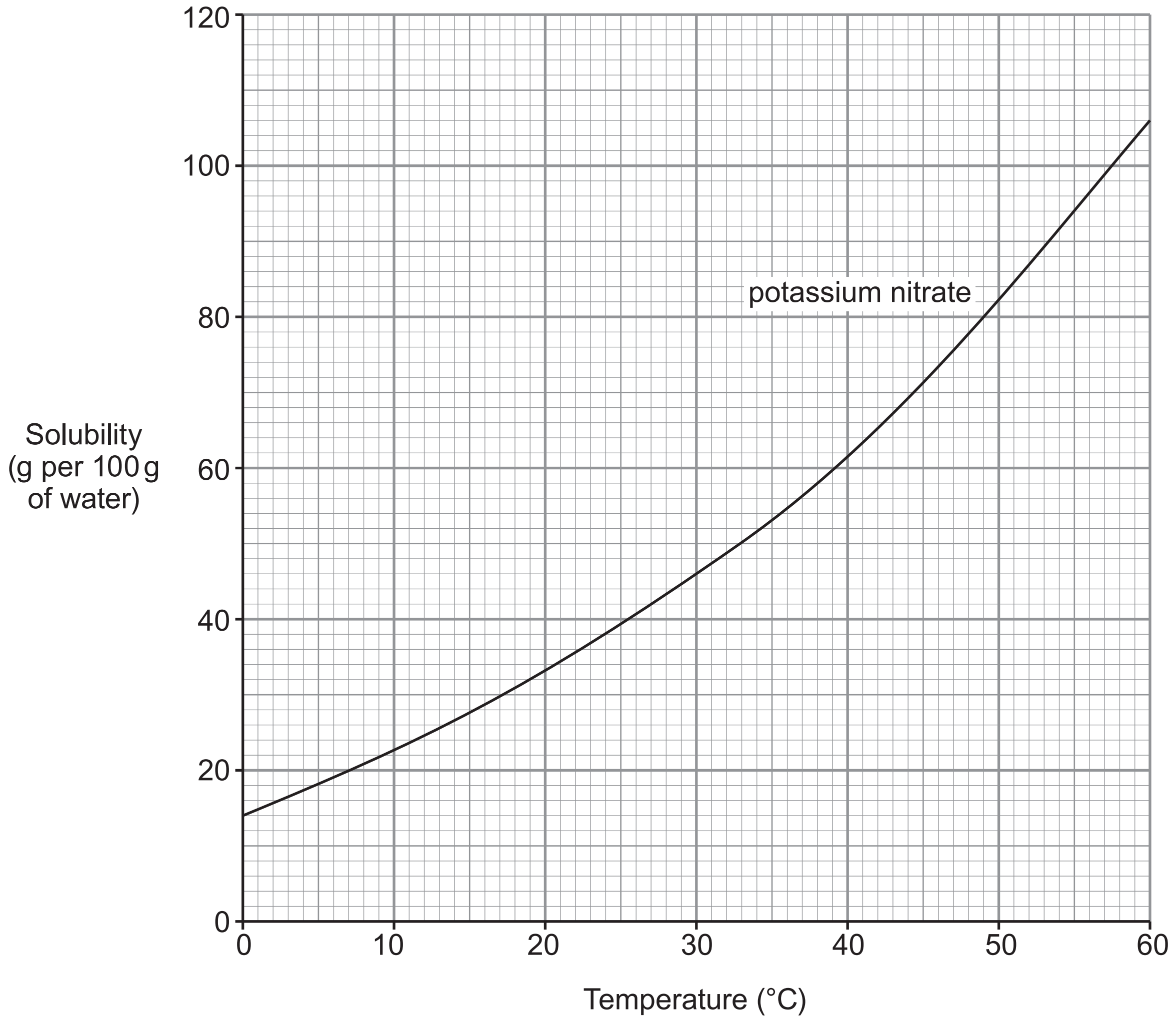
(b) The graph below shows the solubility of potassium nitrate at different temperatures.



Use the information in the graph to answer parts (i) and (ii).

(i) 60 g of potassium nitrate was added to 100 g of water at 30 °C. After stirring the mixture some of the potassium nitrate did not dissolve.

Calculate the mass of potassium nitrate which did **not** dissolve. [1]

Mass = ... g

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(ii) On cooling a saturated solution of potassium nitrate containing 100 g of water from 55 °C to a lower temperature, 36 g of solid was formed.

Determine the temperature to which the solution was cooled.

Show your working. [3]

Temperature = ... °C

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